

Arduino 3 LED On/Off Project

Alireza Naderinasab

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1 Project Description

This project demonstrates sequential control of three LEDs using Arduino. Each LED turns ON and OFF one after another with a 1-second delay. This is a basic introduction to digital output control.

2 Code

```
int ledPin_1 = 11;
int ledPin_2 = 12;
int ledPin_3 = 13;

void setup(){
    pinMode(ledPin_1 , OUTPUT);
    pinMode(ledPin_2 , OUTPUT);
    pinMode(ledPin_3 , OUTPUT);
}

void loop(){
    digitalWrite(ledPin_1 , HIGH);
    delay(1000);
    digitalWrite(ledPin_1 , LOW);
    delay(1000);

    digitalWrite(ledPin_2 , HIGH);
    delay(1000);
    digitalWrite(ledPin_2 , LOW);
    delay(1000);
```

```
        digitalWrite(ledPin_3 , HIGH);  
        delay(1000);  
        digitalWrite(ledPin_3 , LOW);  
        delay(1000);  
    }
```

3 Hardware Setup

- Arduino board
- 3 LEDs
- 3 resistors (220Ω)
- Breadboard and wires

4 Output Images

4.1 LED 1



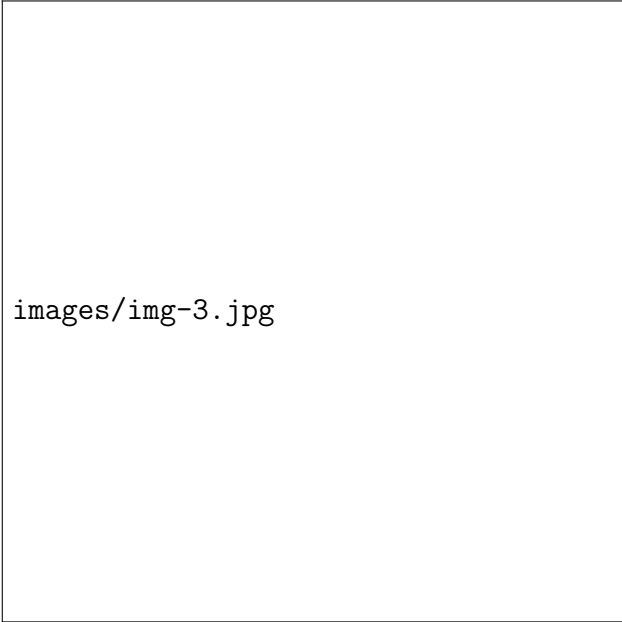
images/img-1.jpg

4.2 LED 2



images/img-2.jpg

4.3 LED 3



images/img-3.jpg

5 Conclusion

This project demonstrates basic sequential digital output control using Arduino. It can be extended for more complex embedded systems.